Institute of Engineering & Management

Department of Electrical and Electronics Engineering

Report on the Five-Day Faculty Development Programme (FDP)

on "FUTURE INNOVATIONS IN QUANTUM VLSI: AI-Driven Approaches in Quantum Computing, and VLSI"

27th – 31st October 2025 (Hybrid Mode)

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1. Introduction

The Department of Electrical and Electronics Engineering, Institute of Engineering & Management (IEM), successfully organized a **Five-Day Faculty Development Programme (FDP)** titled *"FUTURE INNOVATIONS IN QUANTUM VLSI: AI-Driven Approaches in Quantum Computing, and VLSI"* from **27th to 31st October 2025** in **hybrid mode**.

The rapid evolution of technology has led to the convergence of **Artificial Intelligence (AI)**, **Quantum Computing**, and **VLSI (Very Large Scale Integration)** — three domains that are redefining the future of computation and intelligent systems. Traditional VLSI design approaches are now reaching their physical and architectural limits, paving the way for **Quantum VLSI**, where **quantum principles** such as superposition and entanglement are harnessed to perform complex computations with unprecedented efficiency.

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2. Objectives of the FDP

The primary objectives of the programme were:

- Introduce the fundamentals of **Quantum VLSI** and its potential impact on future chip design.
- Demonstrate how **AI and Machine Learning** can enhance design automation, optimization, and fault detection in VLSI systems.
- Explore the role of **quantum algorithms and architectures** in revolutionizing computation and communication technologies.
- Encourage **interdisciplinary research and innovation** at the intersection of AI, Quantum Computing, and VLSI.

- Empower faculty to **integrate these advanced concepts into teaching, research, and project development** for future-ready engineering education.

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3. Programme Overview

The five-day FDP consisted of **expert sessions by eminent speakers** from reputed institutes and industries across India and abroad. Each session included interactive discussions and Q&A segments to engage participants effectively.

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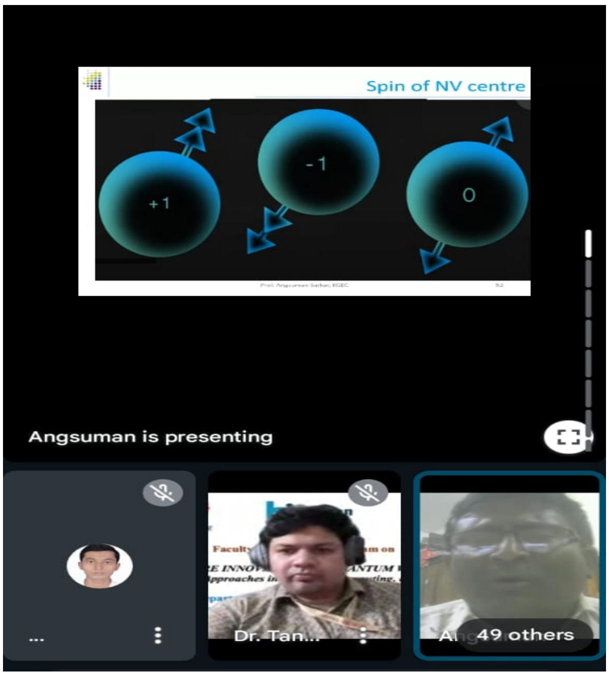
4. Day-wise Summary of Sessions

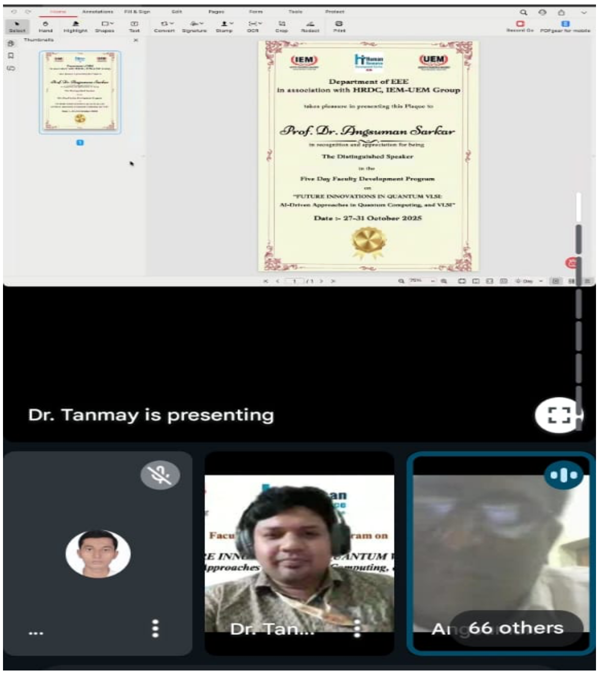
Day 1 – 27th October 2025

Speaker: Prof. Dr. Angsuman Sarkar, Professor & HOD, Department of Electronics and Communication Engineering, Kalyani Government Engineering College

Topic: *Quantum VLSI and Quantum Biosensing: Opportunities, Trends and Challenges*

The FDP commenced with an inaugural session, followed by a lecture by Dr. Sarkar. Quantum VLSI and Quantum Biosensing represent the next frontier in nanoscale technology and intelligent systems. They combine quantum mechanics with advanced VLSI design and biosensing for ultra-fast computation and precise detection. Emerging trends focus on quantum transistor architectures, qubit integration, and quantum-enabled diagnostics. These innovations offer transformative opportunities in healthcare, communication, and computing. However, challenges in scalability, stability, and fabrication remain key areas for future research.





Day 2 – 28th October 2025

Speaker: Mr. Ayan Bhattacharya, Senior ADAS Engineer, Mercedes Benz R&D India, Bangalore

Topic: *AI in Automotive: Best Use Cases, Benefits, & Cautions*

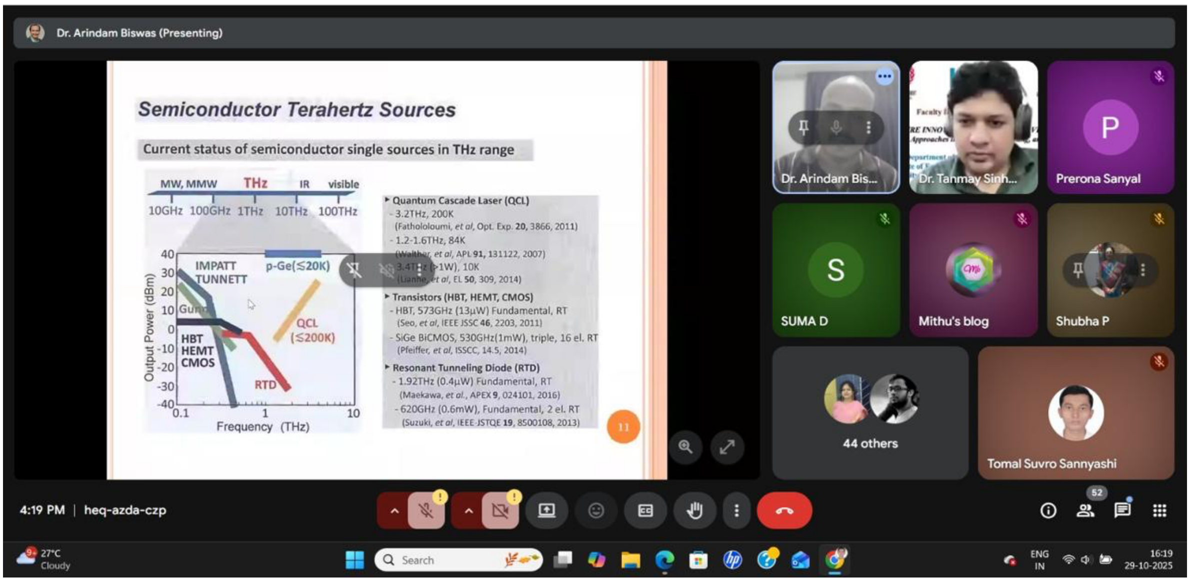
Artificial Intelligence (AI) is transforming the automotive industry through innovations in safety, automation, and efficiency. Key applications include autonomous driving, predictive maintenance, driver monitoring, and smart navigation. AI enhances vehicle performance, reduces human error, and enables personalized driving experiences. However, challenges like data privacy, system reliability, and ethical concerns must be addressed. Balancing innovation with safety ensures a responsible and intelligent future for automotive technology.

Day 3 – 29th October 2025

Speaker: Prof. Dr. Arindam Biswas, Associate Professor & HOD, Department of Computer Science & Engineering, Kazi Nazrul University

Topic: *Application of ANN for Optimizing VLSI-Based IMPATT Oscillator*

Artificial Neural Networks (ANNs) offer intelligent solutions for optimizing VLSI-based IMPATT oscillators used in high-frequency applications. By learning complex nonlinear relationships, ANNs enhance design accuracy and performance prediction. They enable efficient tuning of parameters such as frequency, power output, and efficiency. Simulation-driven ANN models reduce design time and improve circuit reliability. Thus, ANN-based optimization represents a smart and adaptive approach to next-generation IMPATT oscillator design.

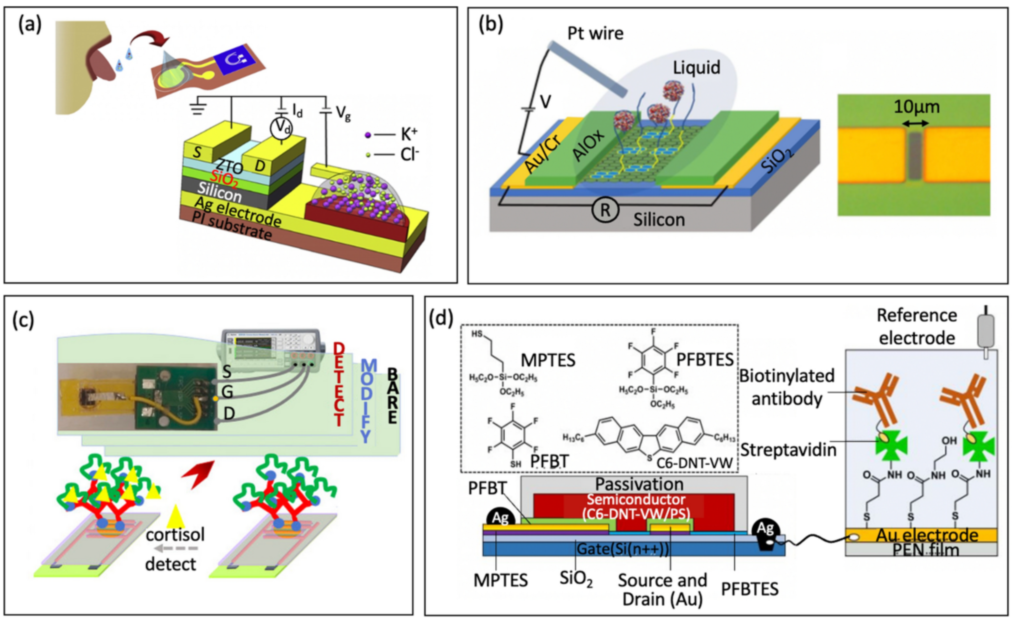


Day 4 – 30th October 2025

Speaker: Prof. Dr. Manodipan Sahoo, Associate Professor, Department of Electronics Engineering, IIT(ISM) Dhanbad

Topic: *Modeling and Simulation of FET based Biosensors*

Field Effect Transistor (FET)-based biosensors are emerging as powerful tools for rapid and sensitive bio-detection. Modeling and simulation play a vital role in understanding their electrical and biochemical interactions. These approaches enable accurate prediction of sensor behavior under varying biological conditions. Through computational analysis, device optimization and performance enhancement become more efficient. Thus, modeling and simulation serve as the foundation for designing next-generation FET biosensing technologies.



Day 5 – 31st October 2025

Speaker: Mr. Bikram Biswas, Analog Mixed Signal Engineer, Samsung Semiconductor India Research, Bangalore
Topic: *AI Applications in the VLSI Design: Bridging Intelligence and Silicon*

The final day's session explored with Artificial Intelligence (AI) is revolutionizing the VLSI design landscape by enabling smarter, faster, and more efficient chip development. AI-driven algorithms optimize circuit layout, power consumption, and fault prediction with remarkable precision. By integrating intelligence into silicon, designers can achieve autonomous design automation and enhanced performance. This fusion accelerates innovation in semiconductor technology, paving the way for next-generation intelligent hardware. Thus, AI serves as a transformative bridge between computational intelligence and advanced VLSI systems.

5. Participation and Engagement

The FDP witnessed enthusiastic participation from **faculty members, research scholars, and industry professionals** from across the country. Interactive Q&A sessions, online discussions, and feedback interactions created an engaging and productive learning environment. The sessions were conducted via the Zoom platform, ensuring accessibility and flexibility for participants.

6. Outcomes of the Programme

- Gain a comprehensive understanding of Quantum VLSI concepts and their significance in next-generation computing.

- Explore the role of Artificial Intelligence in Quantum Computing and VLSI design automation**.**
- Apply AI-driven optimization techniques for quantum circuit modeling, simulation, and performance enhancement.
- Understand emerging research trends, tools, and challenges in integrating AI with Quantum and VLSI technologies.
- Develop the ability to incorporate advanced concepts into teaching, research, and project-based learning for academic and industrial growth.

All participants received **E-certificates** upon successful completion of the programme.

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7. Acknowledgements

The Department of Electrical and Electronics Engineering extends heartfelt gratitude to all distinguished speakers for their valuable time and contributions. Special thanks to the organizing committee members, technical team, and participants for making this FDP a grand success.

The department also acknowledges the support and encouragement provided by the management of the Institute of Engineering & Management, Kolkata which played a vital role in the smooth execution of this event.

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8. Conclusion

The **5-Day Faculty Development Programme (FDP)** on *“FUTURE INNOVATIONS IN QUANTUM VLSI: AI-Driven Approaches in Quantum Computing, and VLSI”* concluded successfully on 31st October 2025. The programme enriched participants with a blend of theoretical insights and practical applications, aligning with the institute’s continuous endeavor to foster research and innovation among faculty members.